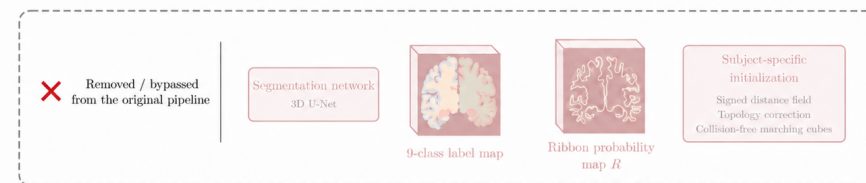
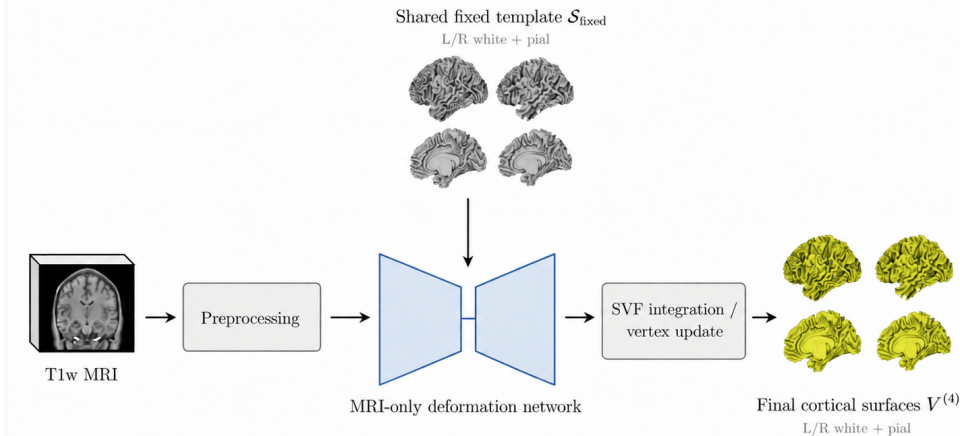


# Fixed-Template SimCortex: Investigating the Impact of Initial Surface Geometry on Cortical Reconstruction

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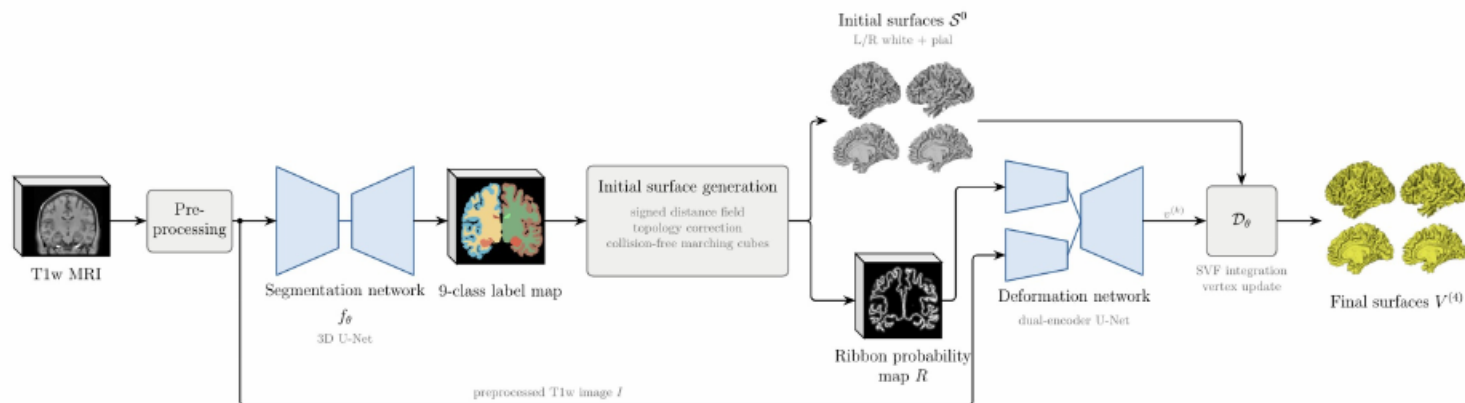
Research question

**How does the geometry of a shared fixed initial surface affect reconstruction accuracy and structural validity?**

## Original SimCortex v2: subject-specific initialization

BACKGROUND

The v2 pipeline generates subject-specific initial surfaces and ribbon information before joint deformation.



### ORIGINAL V2

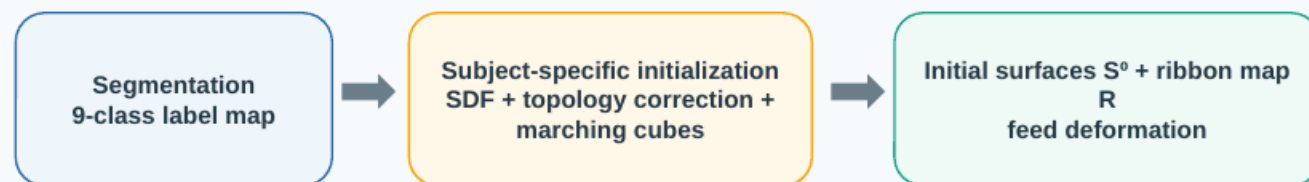
#### What happens before deformation?

1. Segmentation network predicts a 9-class cortical label map.
2. Initial-surface generation constructs subject-specific LH/RH white + pial surfaces.
3. A ribbon probability map is also generated for deformation.
  - And after this, the deformation network then jointly deforms the four surfaces and ribbon information toward the final cortical reconstruction.

#### KEY POINT

**The initialization is regenerated for every subject.**

### SimCortex v2 before deformation



# Why replace subject-specific initialization?

Question

## 1 BOTTLENECK Subject-specific initialization repeats for every MRI

Segmentation and surface generation must run before deformation for each new subject.

## 2 IDEA Precompute the starting geometry once

Reuse one shared set of LH/RH white + pial surfaces instead of rebuilding  $S^0$  for every case.

## 3 QUESTION How much performance can the simpler pipeline recover?

Measure the trade-off in geometry, collision, self-intersection, thickness, and runtime.

### RUNTIME MOTIVATION

Original SimCortex v2

**165.0 s** per subject

Initial-surface generation alone

**112.6 s**

68.2% of total runtime

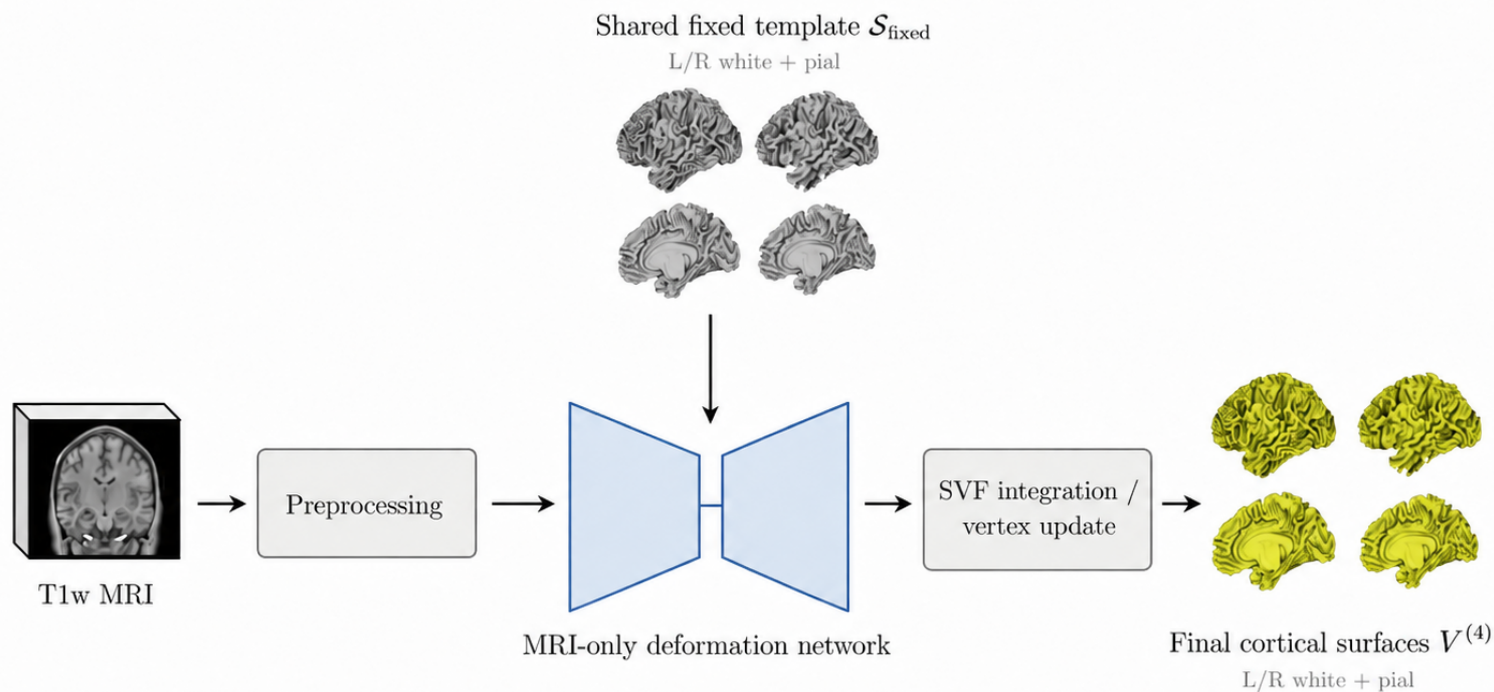
The goal is to remove the per-subject initialization branch, not merely accelerate it.

Research question: can a precomputed shared template preserve useful accuracy and validity while removing the dominant initialization cost?

# My modification: a shared fixed initial template

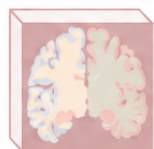
MY PROJECT

At inference, all subjects reuse the same four surfaces; segmentation and subject-specific surface generation are bypassed.

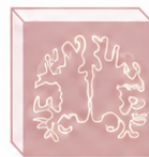


Removed / bypassed  
from the original pipeline

Segmentation network  
3D U-Net



9-class label map



Ribbon probability  
map  $R$

Subject-specific  
initialization  
Signed distance field  
Topology correction  
Collision-free marching cubes

REMOVED

## Per-subject initialization branch

No per-subject segmentation, label map, or initial-surface generation at inference.

ADDED

## Shared fixed initial surfaces

The same four LH/RH white + pial surfaces are reused for every MRI.

CONTROLLED

## Across fixed-template experiments

Same deformation framework, training protocol, loss, checkpoint rule, and evaluation.

# Experiment pipeline and data

DESIGN

Each template follows the same train -> checkpoint -> inference -> evaluation path.



Training data

**HCP-YA + OASIS-1 | MRI only**

External benchmark

**14 datasets × 40 subjects = 560 cases**

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| SURFACES / METHOD                                                              | FCL PAIRS / METHOD | CHECKPOINT RULE                                                                                                                             |
|--------------------------------------------------------------------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2,240</b>                                                                   | <b>3,360</b>       | <b>Best RMSE</b>                                                                                                                            |
| Same cohort, same coordinate conversion, same metrics, same exact FCL backend. |                    | 6 PAIRS<br>LH white ↔ LH pial<br>RH white ↔ RH pial<br>LH pial ↔ RH pial<br>LH white ↔ RH white<br>LH white ↔ RH pial<br>RH white ↔ LH pial |

Evaluation cohort: sample40\_clean\_manifest\_v1. Completed arms use 560/560 cases with strict QC and zero missing cases.

# Five methods in the final comparison

MATRIX

One full-pipeline reference plus four fixed-template experiments. The same names are used consistently throughout the talk.

## Simcortex V2

Original full pipeline, segmentation +  
subject-specific initial surfaces

full-pipeline performance reference

## Sphere

Clean nested icospheres, subdivision-7  
synthetic white/pial shells

same connectivity  
trained + evaluated

## Random

OASIS-1 train split  
sub-0447, seed 2025 • unsmoothed  
anatomical initialization

reproducible random-anatomy control

## Original150k

Anatomical sub-298051, Taubin-smoothed  
to 150k

smooth anatomical fixed-template control

## Edge-only remeshed curl = 1

Curvature-aware Edge-only, LCC clean •  
target  $\approx 0.618$  mm • curvature adaptation 1

final remeshed anatomical candidate

# Taubin smoothing: smooth the mesh without continuous shrinkage

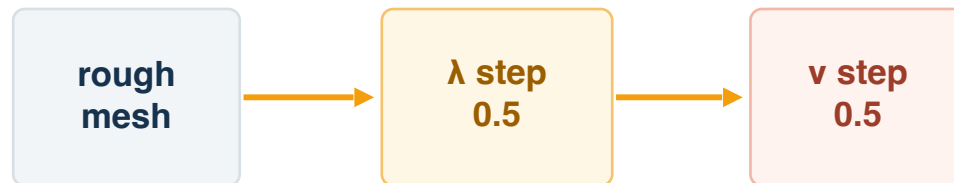
SMOOTHING

A complete  $\lambda/v$  pair is treated as one atomic smoothing block, then structurally audited.

## Why smooth?

- Remove high-frequency mesh irregularities while retaining the large-scale cortical anatomy.
- Avoid the progressive surface shrinkage of plain Laplacian smoothing

### one complete Taubin block

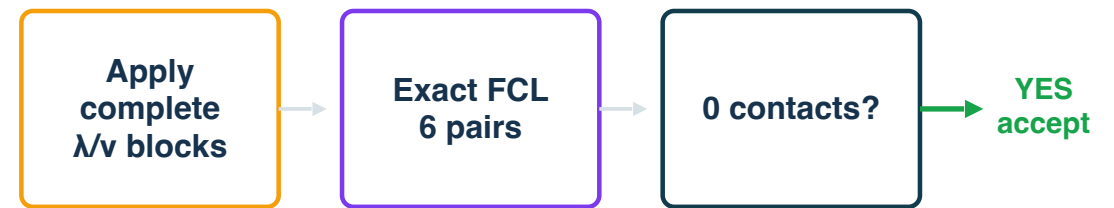


Step-size parameters from Taubin smoothing:

$\lambda$  step: smooths the surface by moving vertices in the Laplacian direction

$v$  step: a corresponding correction step that reduces the shrinkage that occurs in general Laplacian smoothing.

## Collision-aware acceptance



Ex: While doing smoothing iterations from 9000 to 9500: 9500- 9000 =500 is « block ». And if we found the collision while doing this iteration, correct the collision by using the left method, and halve the « block »: 500 to 250, 9000 to 9250.

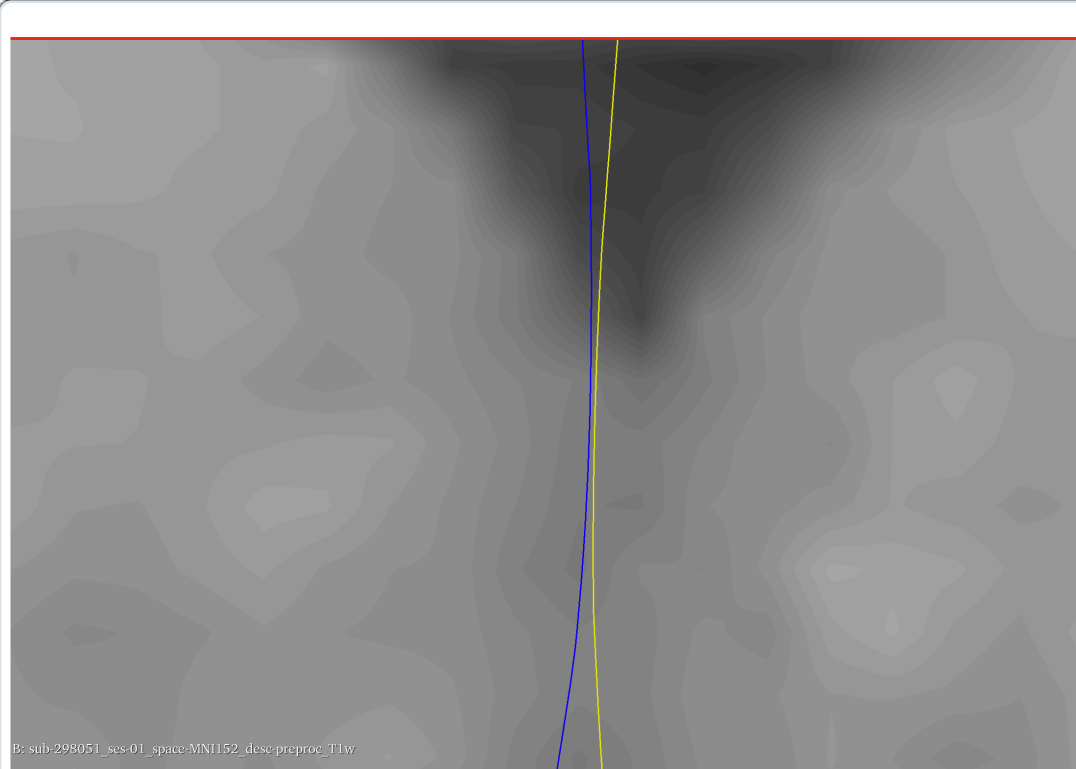
NO

**Rollback to last safe state -> protect contact neighbourhood -> retry smoothing elsewhere**

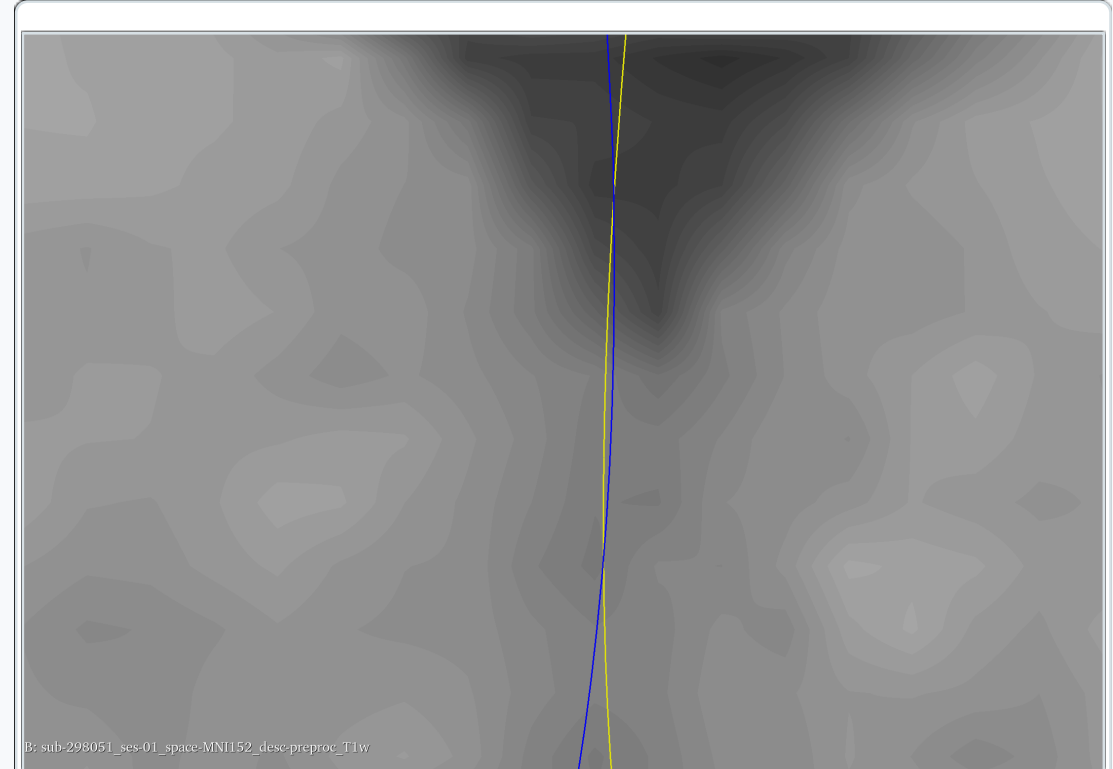
# Why stop at 150,000 smoothing iterations?

**STOPPING RULE**

150k is an empirical stopping point for this anatomical template, not a universal Taubin constant.



**150k: narrow pial separation is still preserved in the selected working state.**



**200k : additional smoothing closes the gap and pial-pial contact / crossing appears.**

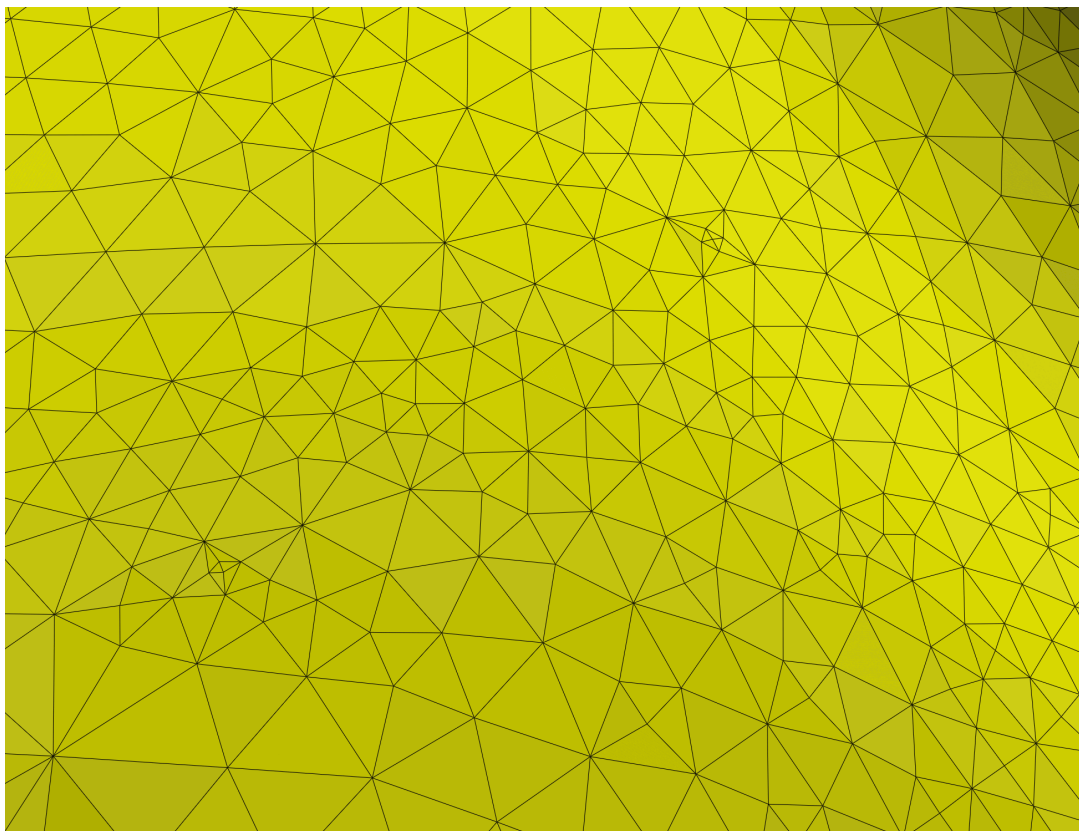
**More smoothing is not automatically safer. Stop before geometric regularization destroys shell separation. The accepted state must remain smooth enough and collision-free.**

*Replace the red placeholder with the final 200k pial-pial Slicer screenshot before the talk.*

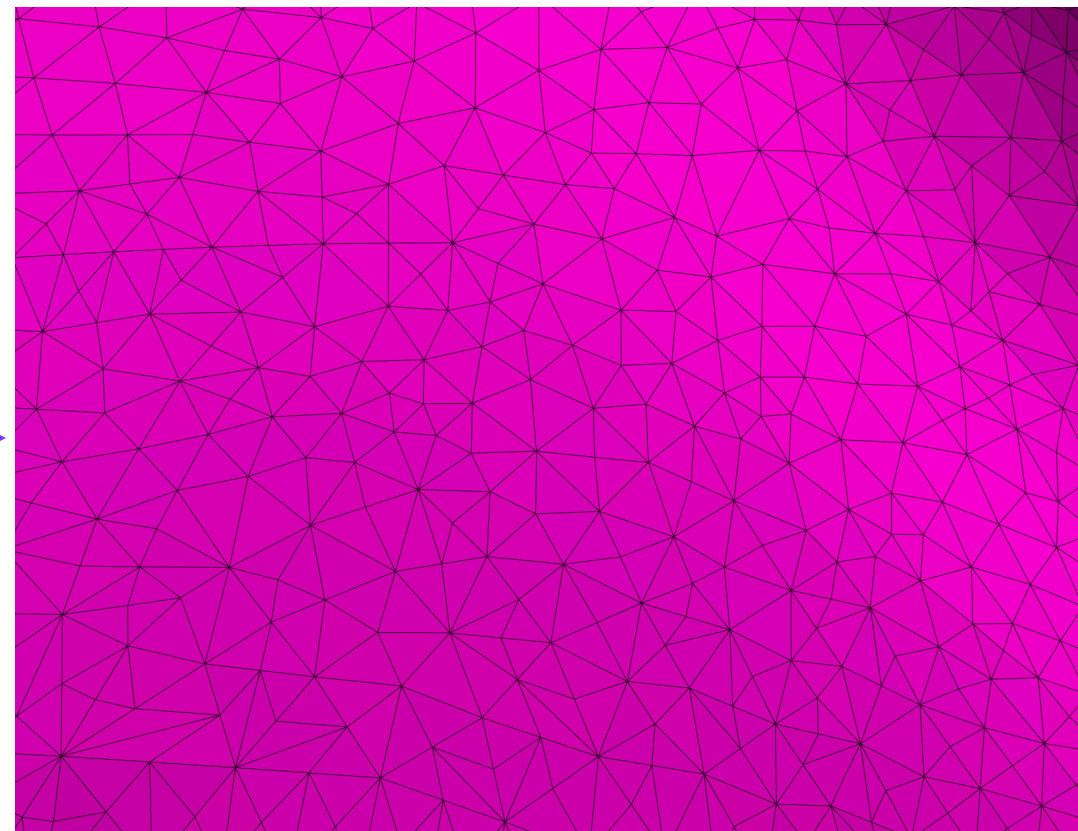
# Why remesh the anatomical 150k template?

3D SLICER

The goal is not to erase folds; it is to regularize local graph sampling with minimal geometric drift.



Original 150k: irregular local edge lengths



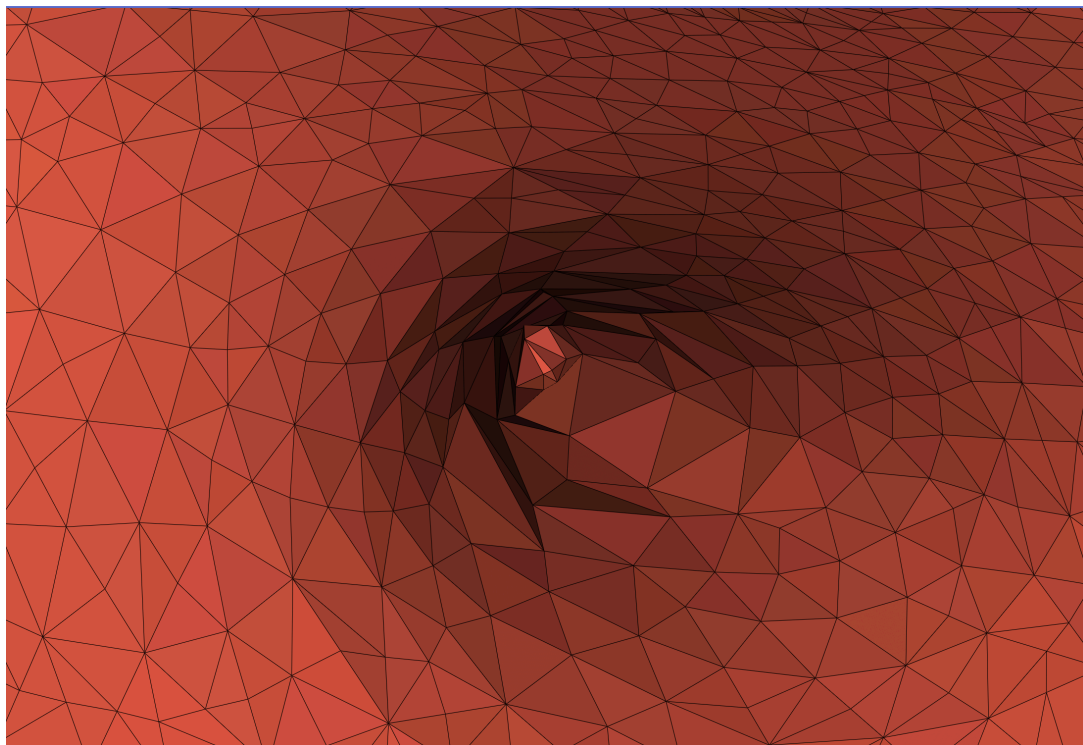
Edge-only: edge lengths move toward  $\approx 0.618$  mm

Same folds, more regular sampling -> better-conditioned graph initialization

# curv=0 vs curv=1: where do we spend triangles?

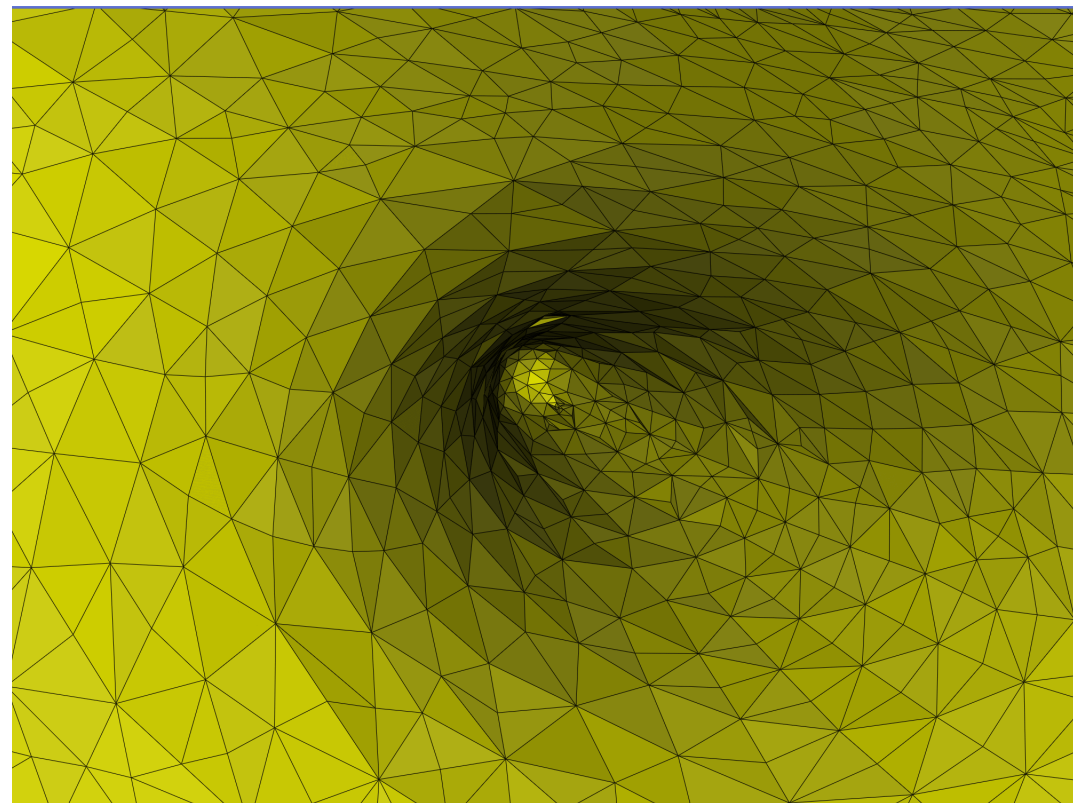
CURVATURE

Curvature adaptation changes the sampling objective, not the overall anatomical target.



curv=0: more uniform edge-length objective

curv=0: curvature does not modify the target edge distribution.



curv=1: retain more resolution in high-curvature folds

curv=1: high-curvature regions receive relatively finer sampling.

Curvature adaptation = 1 gives shorter target edges in strongly curved regions, so folds receive more vertices. This may improve local geometric fit, but the later SIF increase is treated as a hypothesis to investigate, not a proven causal effect.

# Controlled training for the four fixed-template experiments

TRAINING

Sphere, Random, Original Smooth 150k, and Remeshed Curv=1 use the same fixed-template MRI-only training framework.

Data

**HCP-YA + OASIS-1**  
**MRI only**

Budget

**600 epochs**  
**validation every 5 epochs**

Optimization

**LR = 1e-4**  
**weight decay = 1e-4**

Sampling

**180k train points**  
**150k validation points**

Model

**Same Surfdeform**  
**Same architecture/losses**

Checkpoint

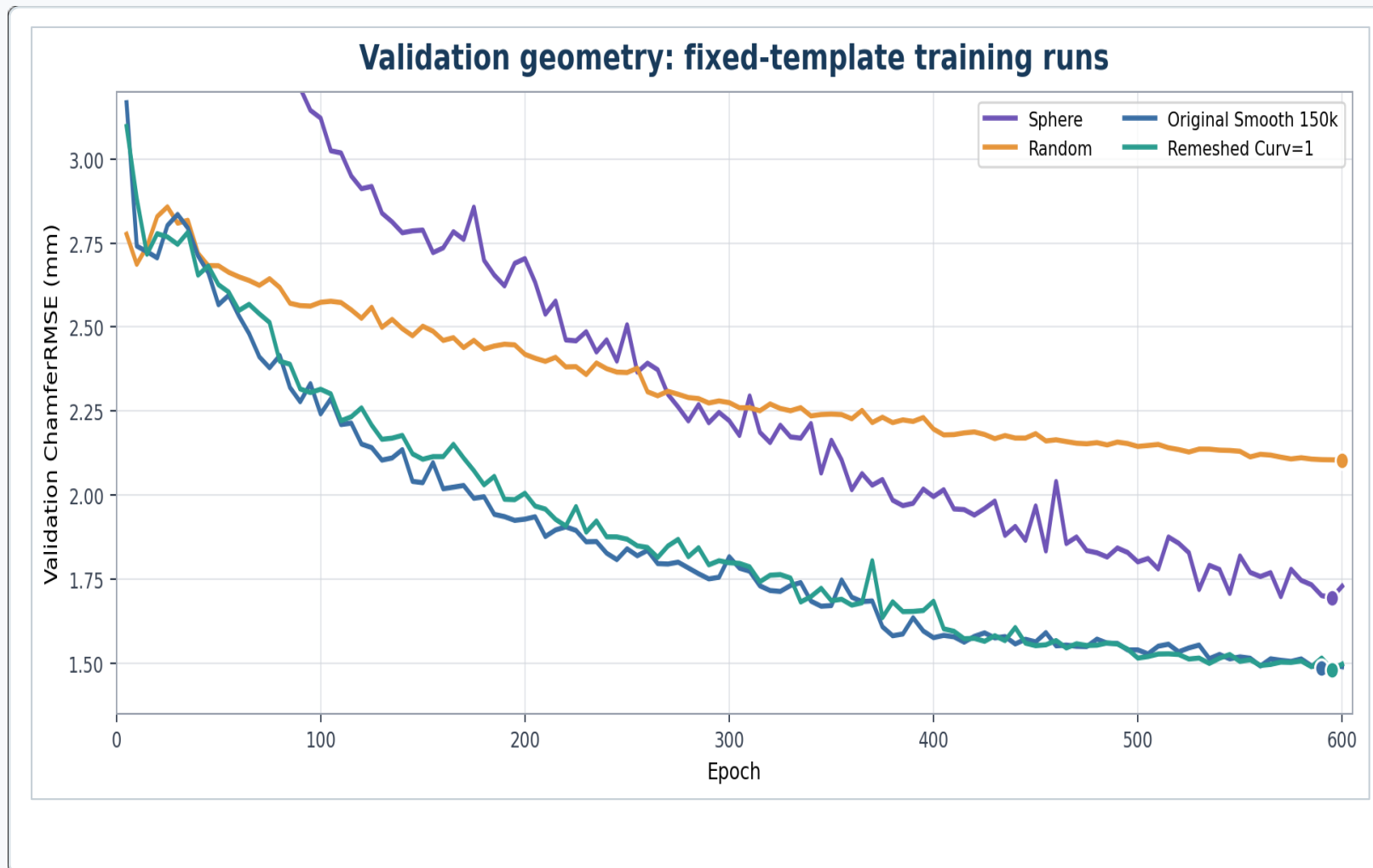
**best validation**  
**Chamfer RMSE**

SimCortex v2 is excluded here because it is the original full pipeline, not the same controlled fixed-template training experiment.

# TensorBoard: validation Chamfer RMSE

TENSORBOARD

Each circle marks that method's best validation ChamferRMSE checkpoint used for external inference.



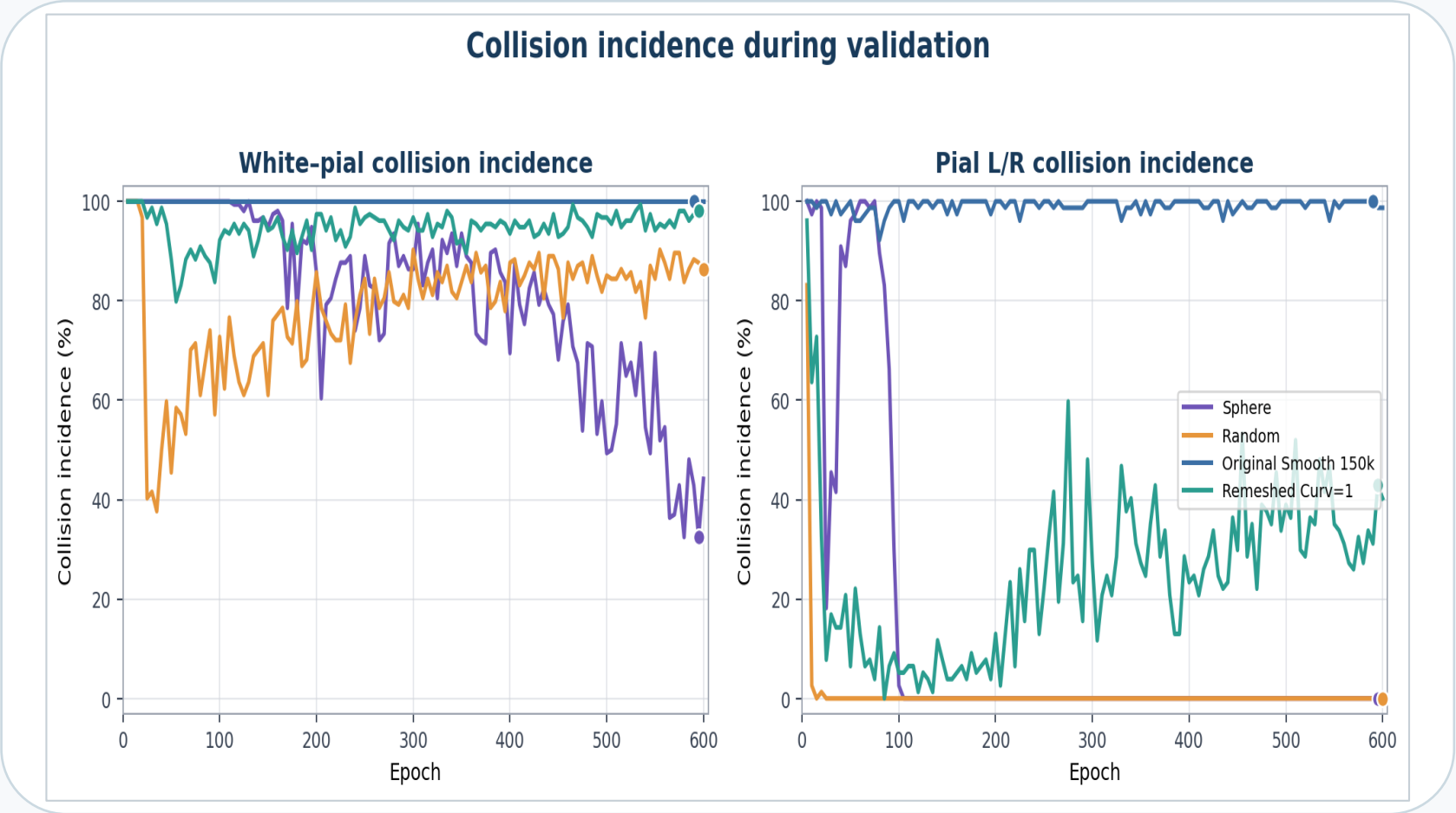
## Best validation RMSE

|              |           |
|--------------|-----------|
| Original150k | 1.4854 mm |
| Curv=1       | 1.4806 mm |
| Sphere       | 1.6944 mm |
| Random       | 2.1039 mm |

# Collision incidence during validation

INCIDENCE

Binary incidence asks whether a validation pair has at least one exact FCL contact; it does not measure how large the contact burden is.



## Best validation RMSE

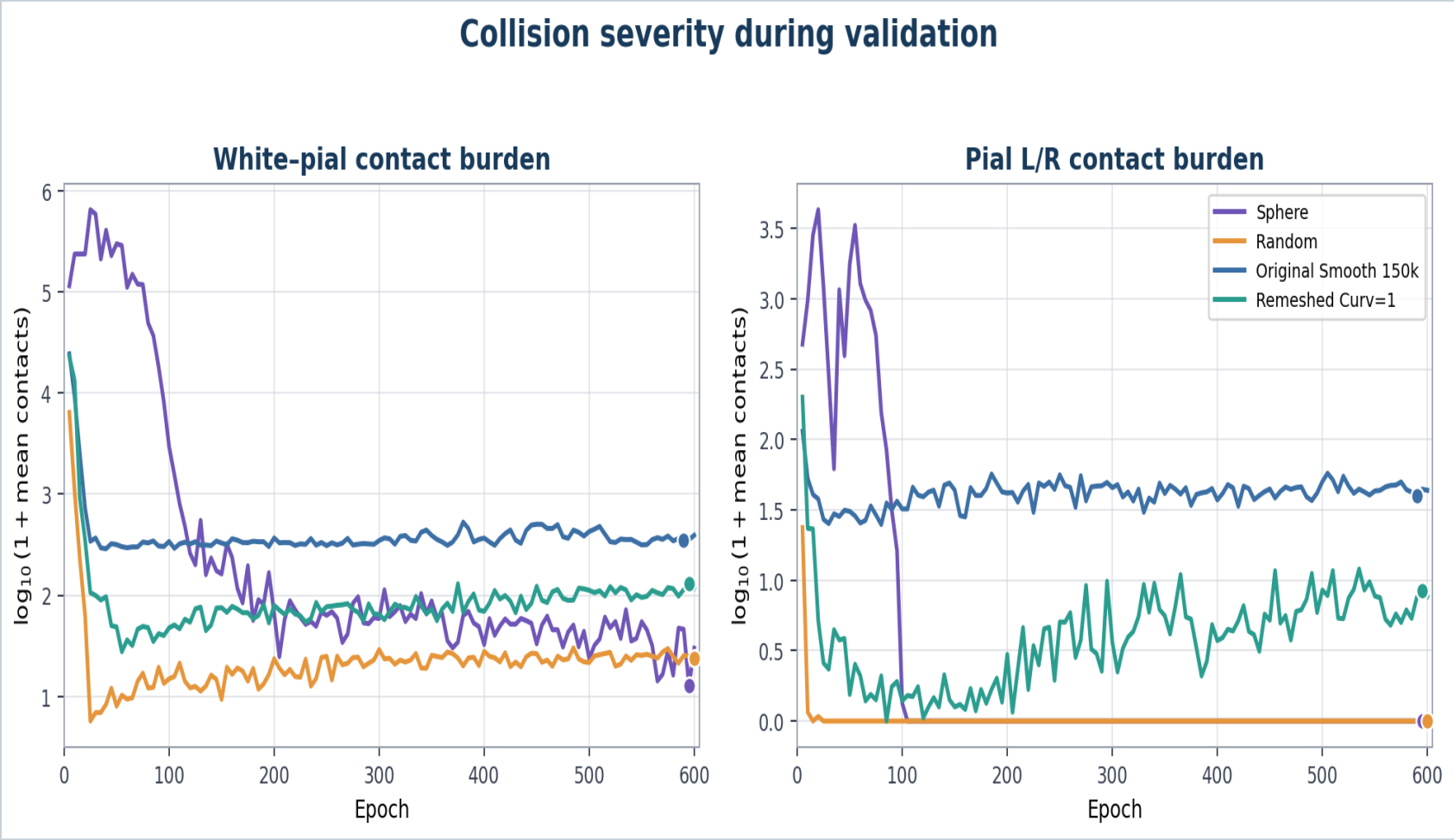
| Methods      | W-P    | Pial L/R |
|--------------|--------|----------|
| Original150k | 100.0% | 100.0%   |
| Curv=1       | 98.1%  | 42.9%    |
| Sphere       | 32.5%  | 0.0%     |
| Random       | 86.4%  | 0.0%     |

Left: % of same-hemisphere white-pial validation pairs with any exact FCL contact. Right: % of cases with a left-right pial contact. Validation is sampled every 5 epochs.

# Collision severity during validation

SEVERITY

Severity uses  $\log_{10}(1 + \text{mean exact FCL contacts among colliding pairs})$ , so distance on the y-axis corresponds to orders of magnitude in contact burden.



## MEAN CONTACTS

| Methods      | W-P   | Pial L/R |
|--------------|-------|----------|
| Original150k | 350.5 | 39.1     |
| Curv=1       | 129.7 | 7.4      |
| Sphere       | 11.8  | 0.0      |
| Random       | 22.8  | 0.0      |

Main training insight: Curv=1 has far less white-pial contact burden than Smooth150k, even though binary incidence remains high.

The log transform keeps very large early contact counts readable.

# External evaluation: what each metric measures

EVALUATION

All metrics are lower-is-better and are computed on the same 560-case benchmark.

## Chamfer L1

mm

symmetric point-cloud surface distance

## ASSD

mm

average symmetric surface distance

## HD90

mm

90th-percentile Hausdorff tail error

## SIF

%

fraction of self-intersecting faces

## Thickness error

mm

absolute cortical-thickness error

## Collision union

%

union of faces colliding with any other shell, mean over 4 surfaces

**QC: 560/560 cases | 2,240 surfaces | 3,360 exact FCL pairs | scannerRAS-compatible predictions | strict problems = 0**

# Geometry across the five final methods

RESULTS

Mean over 560 external cases. Lower is better. SimCortex v2 is the original full-pipeline reference; the other four are fixed-template experiments.

| Method               | Chamfer (mm) | ASSD (mm) | HD90 (mm) |
|----------------------|--------------|-----------|-----------|
| SimCortex v2         | 0.529        | 0.253     | 0.586     |
| Sphere               | 0.944        | 0.768     | 1.815     |
| Random               | 1.186        | 1.023     | 2.304     |
| Original Smooth 150k | 0.848        | 0.653     | 1.551     |
| Remeshed Curv=1      | 0.847        | 0.650     | 1.530     |

## WHAT IT SAYS

- REFERENCE

**SimCortex v2 is clearly best**

The original full pipeline remains the geometry target.
- ANATOMICAL PAIR

**0.653 → 0.650 ASSD**

Curv=1 and Smooth150k are nearly tied on pooled geometry.
- CONTROL RANGE

**Sphere > Random**

Random anatomy is the least accurate fixed-template baseline.

Do not interpret the tiny Smooth150k–Curv=1 pooled difference as statistical superiority until matched-case paired testing is complete.

# Structural validity reveals different failure modes

RESULTS

Mean over 560 external cases. Lower is better. Inter-surface collision and within-surface self-intersection must be read separately.

| Method               | SIF (%) | Thickness err. (mm) | Collision union (%) |
|----------------------|---------|---------------------|---------------------|
| SimCortex v2         | 0.044   | 0.124               | 0.00009             |
| Sphere               | 0.0014  | 0.283               | 0.0074              |
| Random               | 0.131   | 0.902               | 0.0083              |
| Original Smooth 150k | 0.249   | 0.277               | 0.0332              |
| Remeshed Curv=1      | 0.566   | 0.352               | 0.0178              |

## WHAT IT SAYS

- COLLISION**

–46.4% vs Smooth150k

0.0332% → 0.0178% collision union.
- TOPOLOGY COST**

+127% SIF

0.249% → 0.566% within-surface self-intersection.
- THICKNESS**

+26.9% error

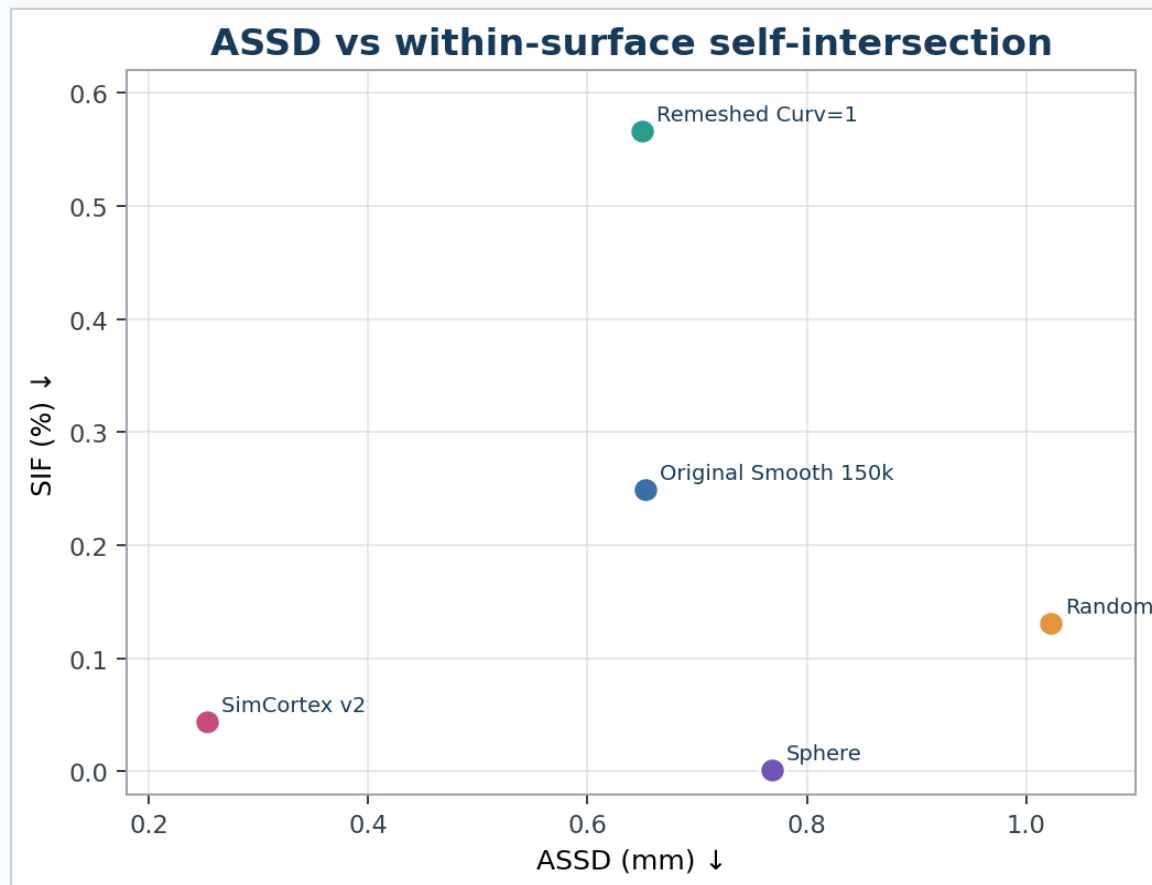
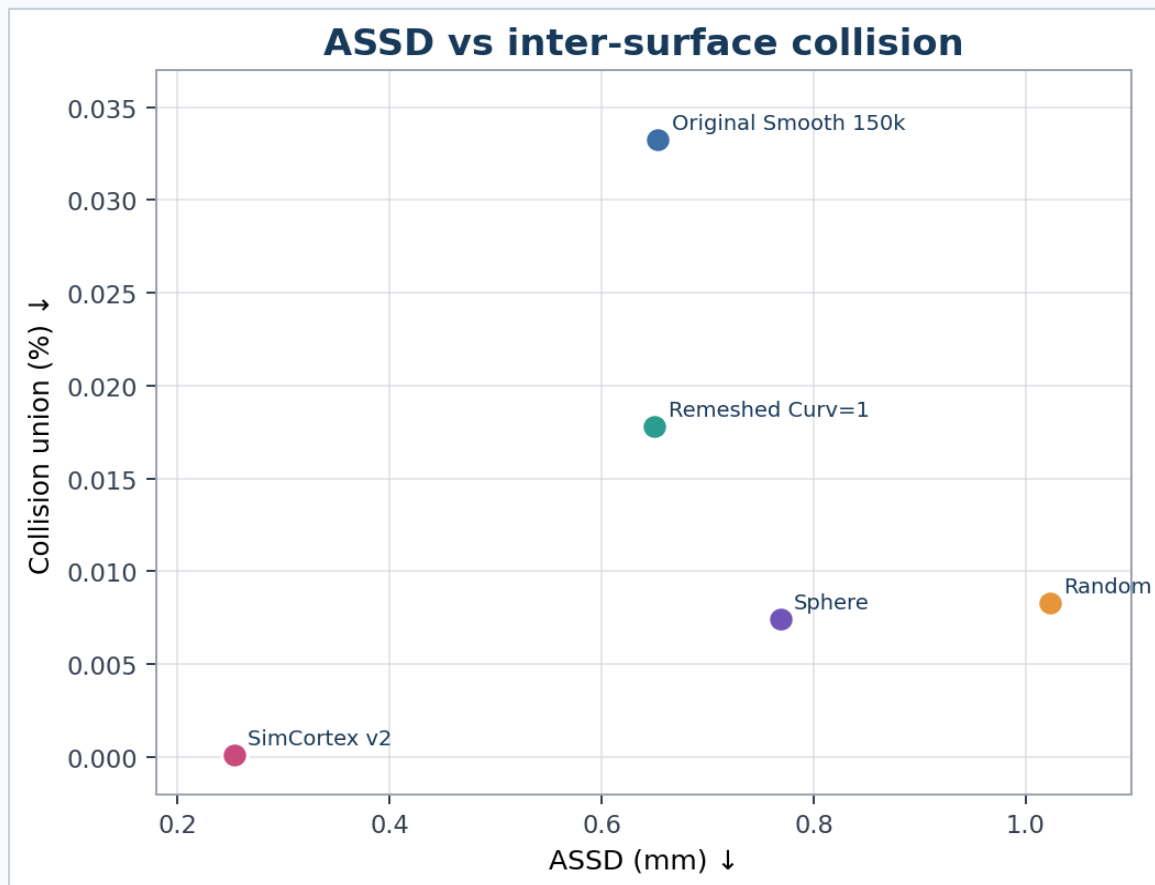
0.277 → 0.352 mm; keep this flagged for a sanity check.

Curv=1 improves inter-surface collision, but that gain does not imply better topology overall because SIF increases strongly.

# ASSD versus collision and self-intersection

TRADE-OFF

Exactly the five final methods. Lower-left is better in both panels; the second panel prevents Curv=1's SIF cost from being hidden.

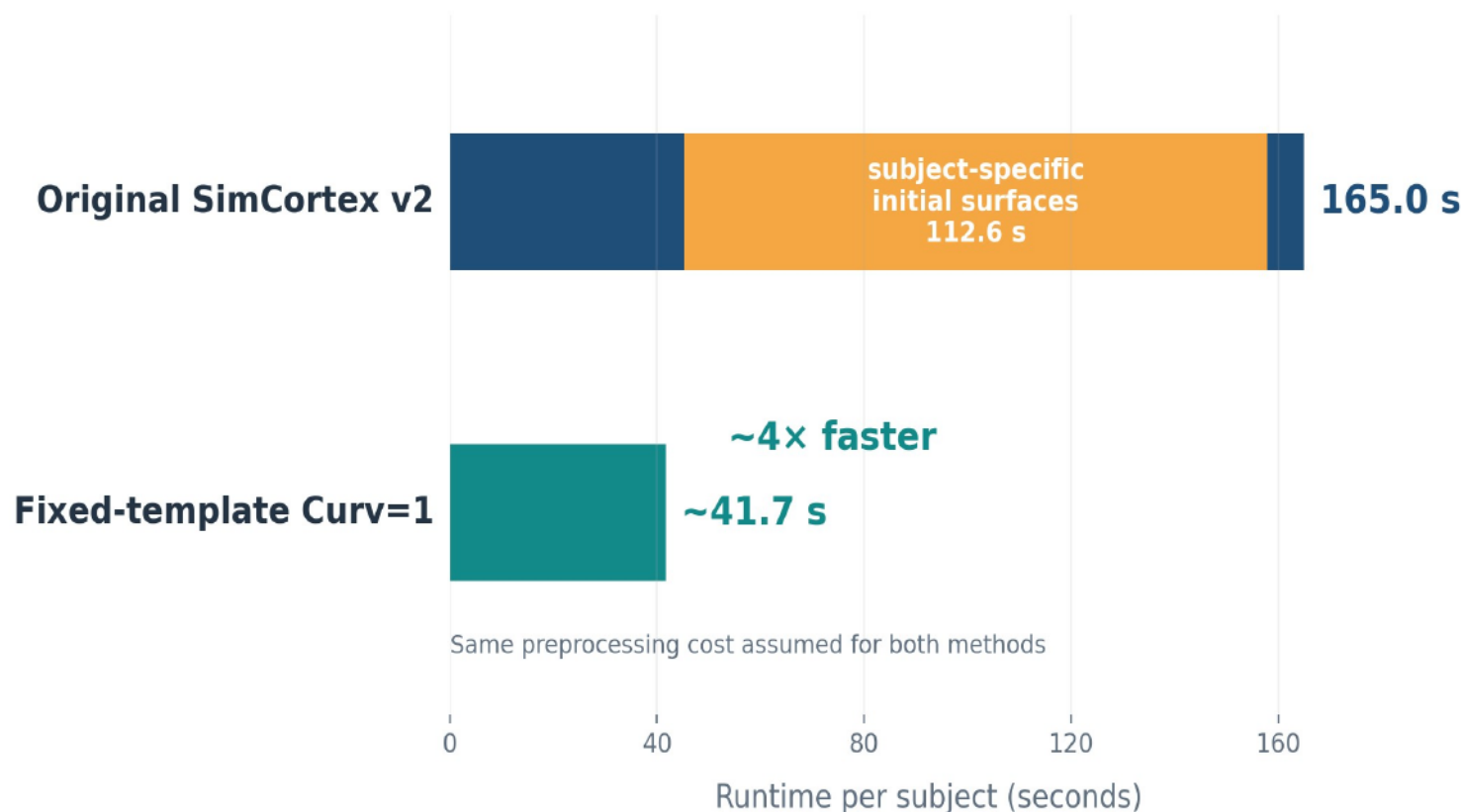


Reading the two plots together: Curv=1 moves left/down relative to Smooth150k in ASSD–collision space, but moves sharply upward in ASSD–SIF space. Sphere stays structurally clean but less accurate; SimCortex v2 remains the performance target.

# Runtime comparison

Original SimCortex v2 vs fixed-template Curv=1

FIGURE 7



RUNTIME

## HOW TO READ

Bars show runtime per subject. Original SimCortex v2 is measured on 40 CNP subjects. The fixed-template value uses the same measured preprocessing cost plus the measured Curv=1 post-preprocessing runtime.

## WHAT IT SAYS

Original SimCortex v2 requires  $165.0 \pm 11.2$  s per subject. Subject-specific initial-surface generation alone takes 112.6 s, or 68.2% of the total. Fixed-template Curv=1 is estimated at ~41.7 s per subject, corresponding to roughly a 4x speed-up.

The speed-up comes from removing subject-specific initialization, not from a faster deformation network.

Original v2: measured over 40 CNP subjects. Fixed Curv=1 end-to-end = 39.46 s preprocessing + 2.27 s measured post-preprocessing runtime.

# Curv=1 improves the fixed-template trade-off

TAKEAWAY

SimCortex v2 remains the performance target; the fixed-template approach is faster, but structural validity still needs work.

## 1 SIMPLIFY

1

### Remove subject-specific initialization

Original v2: 165 s / subject

Initial surfaces alone: 112.6 s

Fixed Curv=1 estimate: ~41.7 s

## 2 PRESERVE

2

### Smooth-anatomy-level geometry

ASSD: 0.650 vs 0.653 mm

Chamfer: 0.847 vs 0.848 mm

Curv=1  $\approx$  Smooth150k on pooled geometry

## 3 IMPROVE

3

### Lower inter-surface collision

Collision union: 0.0332%  $\rightarrow$  0.0178%

$\approx$ 46% reduction vs Smooth150k

Training collision severity also decreases

## 4 FIX

4

### Within-surface self-intersection

SIF: 0.566%

>2 $\times$  Smooth150k

Template audit + SIF analysis still needed

Final interpretation: keep Remeshed Curv=1 as the anatomical fixed-template candidate to refine, but do not claim that it replaces the original SimCortex v2 pipeline.